Filtration with frit technique

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Category: Protocol
Created by: Regina Dietrich

Goal

Filtration of a suspension under argon atmosphere to gain the solid

When should I use a syringe, frit technique or canula technique?

Syringes are used only for the transfer of solvents.

Frit filter technique shoud be used to filter a suspension with the aim to gain the solid. The suspension is transfered from one vessel into the other (with a cannula) through creating static vaccum over the tube of the schlenk line and argon flow from the other side

Cannula (PTFE tube) have a high inside diameter which allows the transfer of suspensions with small particles. Cannulas equipped with a filter can be used to filter a suspension and to gain a homogeneous solution. The suspension is transferred from one flask into the other by pushing it with high pressure of argon.

Prerequisites and preparation

Equipment
Empty schlenk flask, frit, frit transition piece to schlenk,
2x septum
Cannula
Dummy flask

Steps

Step number	Step description
1. Built the setup	The frit is connected with a flask on one side and the transition piece on the other side. The canula is put on both side through a septum. One side is placed on the frit apparature the other side on the dummy flask. The atmosphere is changed to argon. The frit can be carefully heated under vacuum. Argon/vacuum should be opend via the stop cock of the collecting schlenk. The stop cock of the transition piece should be open to the line but the line should be closed. This allows purging of the tubing

2. Connect the cannula with the reaction flask	Under argon flow the canula is placed with the septum in a reaction schlenk. The argon flow is opened to the reaction flask for the whole filtration procedure. the argon flow on the frit apparature comes from the transition piece. The stop cock of the collecting schlenk is closed. The cannula is placed inside the suspension which is stirred. The stop cock of the transition piece is closed (the argon flow of the line is still open to the tubing)
3. Set tube under vacuum	Stop cock of collecting schlenk is still close. The tubing of the collecting schlenk is put under vacuum. The stop cock of the schlenk is closed.
4. Transfer & filtration of the suspension	Carefully the stop cock of the collecting schlenk is opened, creating a under pressure inside the apparature. This allows the suspension to flow through the cannula from the reaction schlenk to/through the frit. The stop cock is closed. *Reminder: During the process of opening the stop cock, the stop cock should not be closed back. If it is closed bevor the vacuum is fully opened, the solution on the frit starts to bubble and can get into the transition piece.*
5. Washing the reaction schlenk	With a purged syringe some solvent is added to the reaction flask. The syringe should be removed afterwards from the septum and should not stuck in it to wait for the next addition of solvent
6. Repeat 2 times step 3-5	
7. Modification of the setup for drying	Argon flow on the reaction schlenk is closed. Argon flow is open on both stop cocks of the filtration apparature. The septum and canulla are removed out and are replaced with a plug. The collecting flask is disconnected from the frit. Collecting flask is closed with a plug and the frit is closed with a dummy flask. Instantely the new frit apparature should be placed under vacuum the remove out the air atmosphere from the dummy schlenk and to dry the filtrated solid

Attached file

Set-Up-Filtration-with-frit-technique.jpeg
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